BOOT / DISARMED
Motors off; sound off

ARM

ARMED / IDLE

AUTO

completed = 0; next behavior = 1
Enable automatic detection

SAMPLE HC-SR04 DISTANCE
At least 70 ms between readings
2 <= distance < 20 cm?

No

Yes

No: reset near-reading count

Increment near-reading count
Three consecutive near readings?

No

Yes

RUN next behavior
Ignore sensor triggers until finished
completed stays unchanged

next = 1

next = 2

next = 3

BEHAVIOR 1: ROAR 1
1.4 s; motors off until roar finishes

BEHAVIOR 2: ROAR 2
2.1 s; motors off until roar finishes

BEHAVIOR 3: ROAR 3
2.8 s; motors off until roar finishes

LEFT PULL
A tug 500 ms + B release 125 ms
Stop each motor at its own deadline

RIGHT PULL
B tug 500 ms + A release 125 ms
Stop each motor at its own deadline

REPEAT 3 CYCLES
A tug 250 ms; B off
80 ms pause

RESET AFTER LEFT
500 ms pause (two 250 ms stages)
B tug 500 ms + A release 125 ms
250 ms final pause

RESET AFTER RIGHT
500 ms pause (two 250 ms stages)
A tug 500 ms + B release 125 ms
250 ms final pause

A release 50 ms + B tug 250 ms
Then 80 ms pause

B release 50 ms; A off
80 ms pause; repeat until 3 cycles

FINISHED: completed += 1
next behavior = 1 -> 2 -> 3 -> 1
500 ms cooldown, then sample again

FROM ANY ACTIVE STATE
STOP: motors/sound off; AUTO off
DISARM / !: also disarm
Aborted behavior does not add a count
ARM + AUTO starts over

EACH MOTOR MOVEMENT
PWM 50 -> 80; rise 100 ms (or shorter)
Stop at the specified elapsed time
PWM controls drive, not measured speed